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CAMPUS WIEN
UNIVERSITY OF
APPLIED SCIENCES

Re-analysis of Alternative Splicing in *SF3B1*-mutated Uveal Melanoma

A Reproduction Study based on Furney et al. (2013) and Harbour et al. (2013)

Course: Workflow Design ILV

Lecturer: Jérémy Le Coz

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1. Introduction

The goal of this report is to reproduce the results found by *Furney et al.* in their work *„SF3B1 mutations are associated with alternative splicing in uveal melanoma*“, particularly regarding their approach to setting up a workflow.

The main focus of this work was to comprehensively understand uveal melanoma genetics. To achieve this, SNP arrays were performed for 12 primary uveal melanoma, identifying around 2000 somatic single nucleotide variants per tumor as well as low levels of aneuploidy. Further, *SF3B1* mutations were identified in three samples and another 15 mutations in a cohort of 105 samples.

While previous studies, such as *Harbour et al. (2013)*, identified recurrent mutations in the splicing factor *SF3B1* at codon 625, they initially failed to find significant global splicing differences between mutated and wild-type samples. *Furney et al. (2013)* re-analyzed these datasets and demonstrated that specific mutations are indeed associated with differential alternative splicing in key genes such as *ABCC5*, *UQCC*, and *ADAM12*.

2. Methods and Workflow of the Original Publications

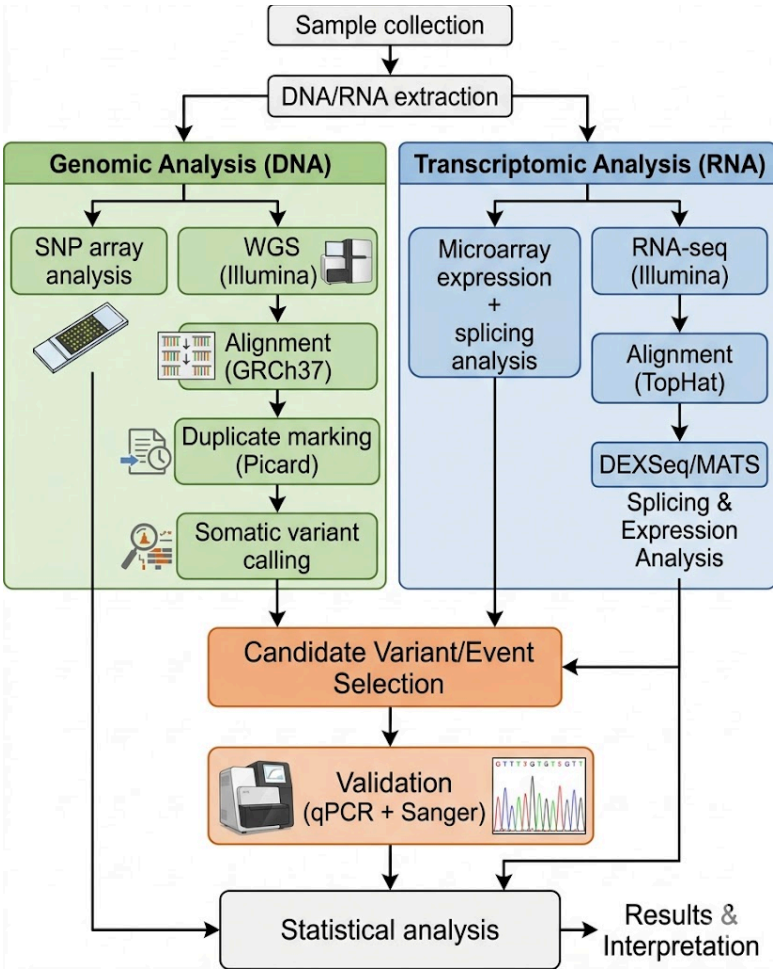
This section details the methodologies employed in the 2013 studies by Furney et al. and Harbour et al. to identify *SF3B1* mutations and their subsequent effect on the transcriptome of uveal melanoma.

The process of sample collection of the original paper is not integral for the workflow automation process of this work. The availability of the data will be presupposed. Further, Whole Genome Sequencing (WGS) is not required for this task.

2.1. Original Experimental Pipeline Overview

To provide a comprehensive view of the discovery phase in 2013, the following sequence was established by the authors:

Workflow: Sample collection → DNA/RNA extraction → WGS (Illumina) → Alignment (GRCh37) → Duplicate marking (Picard) → Somatic variant calling → SNP array analysis → Microarray expression + splicing analysis → RNA-seq (TopHat + DEXSeq/MATS) → Validation (qPCR + Sanger) → Statistical analysis



Created by Gemini (Google, 2026)

2.2. SNP Array Analysis

The studies utilized a combination of Exome Sequencing (Harbour et al.) and Whole-Genome Sequencing (Furney et al.) to identify the *SF3B1* R625 mutation.

To identify large-scale chromosomal aberrations, such as Monosomy 3, SNP array analysis was performed. This provided the necessary genomic background to classify the tumors. In the original workflow, DNA was extracted and hybridized to high-density SNP chips, allowing for the detection of copy number variations (CNVs) and loss of heterozygosity (LOH), which are hallmarks of metastatic uveal melanoma.

2.3. Gene Expression Microarray Workflow

For transcriptome analysis, Furney et al. employed the Affymetrix Human Transcriptome Array 2.0 (HTA2), which is specifically designed to detect exon-exon junctions.

For initial transcriptomic profiling, the studies utilized microarray technology (notably the Affymetrix platform).

1. **Sample collection:** Tumor tissue was obtained via biopsy or enucleation.
2. **RNA extraction:** Total RNA was isolated and quality-checked for integrity.
3. **Hybridization:** The RNA was labeled and hybridized to microarrays (e.g., HTA 2.0) to measure the expression levels of thousands of genes simultaneously, providing the first hint of differential gene expression between mutation groups.

2.4. RNA-seq Analysis Pipeline

The bioinformatic approach differed significantly, which explains the discrepancy in their initial findings:

Harbour et al. focused on global splice donor/acceptor retention and manual analysis of a limited gene set, finding no significant abnormalities.

Furney et al. utilized advanced algorithms for differential exon usage:

- **Alignment:** Reads were mapped using TopHat.
- **Splicing Analysis:** Applied DEXSeq and MATS to identify alternative splicing events with a significance threshold of FDR < 0.1.
- **Validation:** Findings were validated using qRT-PCR for eight candidate genes.

External Data Integration

Preprocessing: Trim reads to 99 bp

2.5. Targeted Validation and Mutation Analysis

To ensure the accuracy of the high-throughput screenings, targeted validation was a crucial step in the original workflow.

Mutation Validation (Targeted Sequencing): The researchers focused on a specific panel of known driver genes including *SF3B1*, *GNAQ*, *GNA11*, and *BAP1*. Mutations in these genes are hallmark features of uveal melanoma and help categorize the tumors into different prognostic groups.

Methods: DNA regions of interest were targeted via PCR amplification followed by Sanger sequencing.

Analysis & Visualization: The sequencing data was processed and visualized using Sequencher Software, allowing for the precise identification of point mutations (especially at codon 625 for *SF3B1*).

Expression Validation: Additionally, qPCR was used to verify the expression levels of candidate genes like *ABCC5* and *ADAM12* that showed significant alternative splicing in the RNA-seq data.

2.6. Statistical Analysis

The significance of the findings was assessed using a rigorous statistical framework.

Survival Analysis: Kaplan-Meier curves and Log-rank tests were used to correlate *SF3B1* status with patient prognosis.

Association Tests: Chi-square and Fisher's exact tests were applied to categorical data.

Significance Threshold: A p-value of < 0.05 (or FDR < 0.1 for high-throughput data) was maintained as the threshold for statistical significance.

3. Modernized Workflow Design

To reproduce the findings of Furney et al. (2013) and Harbour et al. (2013), we evaluated the original methodologies against current bioinformatics standards for 2026. The original papers relied on tools that are now considered suboptimal or deprecated.

3.1. Technology Comparison: 2013 vs. 2026

The following table summarizes the transition from the legacy tools used in the original studies to the modernized pipeline implemented in our project:

Analysis Step	Original Tool (2013)	Modern Equivalent (2026)
Read Alignment	TopHat	HISAT2
Splicing Analysis	MATS / DEXSeq	DEXSeq + rMATS
Workflow Engine	Manual Scripts	Nextflow
Quality Control	Manual / Expression Console	FastQC & MultiQC
Data Handling	Picard	Picard + Samtools

Analysis Step	Original Tool (2013)	Modern Equivalent (2026)
Read Alignment	TopHat	HISAT2
Splicing Analysis	MATS / DEXSeq	DEXSeq + rMATS
Workflow Engine	Manual Scripts	Nextflow
Quality Control	Manual / Expression Console	FastQC & MultiQC
Visualization	GenoSplice EASANA	Sashimi Plots (not applied in this work)

(Furney et al., 2013; Harbour et al., 2013)

3.2. Rationale for Tool Upgrade

HISAT2 is a successor to the deprecated TopHat by the same research group (Kim et al.). Both are splice-aware, HISAT2 is faster and more accurate. It allows better identification of the cryptic splicing events in genes like *CRNDE* and *ABCC5*.

rMATS-turbo offers a more robust Bayesian framework specifically for "Skipped Exon" events, it allows us to re-evaluate the raw data from both studies with a unified, high-performance statistical approach to resolve the discrepancies between the original findings.

Nextflow ensures full reproducibility and scalability of the pipeline and its results.

MultiQC enables the aggregation of QC metrics from all patients into a single interactive report.

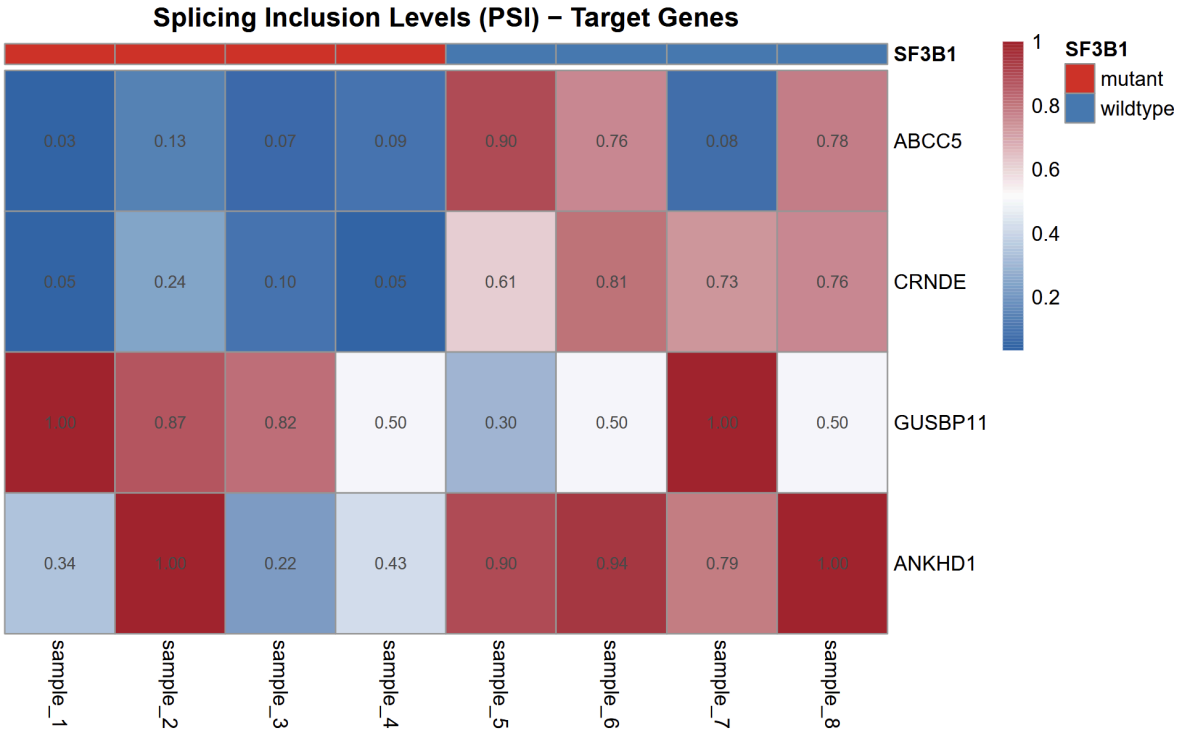
Samtools is a standardized, high-performance tool for modern BAM/SAM file manipulation.

4. Reproduced Results / Report

The reproduction of the original study began with the processing of raw RNA-seq datasets through our modernized pipeline. The initial phase focused on the alignment of sequencing reads to the GRCh37 human reference genome. By

utilizing the HISAT2 aligner on the university's server infrastructure, we achieved a high mapping rate across the patient cohorts. We performed several optimization iterations to refine the alignment parameters, ensuring that the resulting BAM files were correctly coordinate-sorted and indexed for downstream analysis.

Following the successful alignment, the workflow proceeded to the statistical detection of differential splicing events. We applied both rMATS and DEXSeq to quantify exon usage and identify significant splicing variations between *SF3B1*-mutated and wild-type samples. This computational step was designed to specifically validate the alternative splicing patterns in candidate genes such as *ABCC5*, *UQCC*, and *ADAM12*, which were central to the findings of Furney et al. (2013). The pipeline successfully generated the necessary significance values and inclusion levels, providing the basis for the final visualization of splice junctions.



In this heatmap the splicing inclusion levels (PSI - Percent Spliced In) for four target genes across eight samples. The results are grouped by SF3B1 mutation status.

The sample groups consist of:

- samples 1-4: SF3B1 mutant (red header)
- samples 5-8: SF3B1 wildtype (blue header)

ABCC5 shows the clearest pattern in the heatmap. While mutant samples show very low PSI (0.03 - 0.13), most wildtype samples show high PSI (0.76-0.90). The mutation in SF3B1 suppresses the inclusion of exons.

CRNDE shows a similar trend to ABCC5, whereas mutant samples show low PSI (0.05 - 0.24) and wildtype samples show higher PSI (0.61 - 0.81). The difference is visibly consistent, which suggests that SF3B1 mutation also reduces exon inclusion in this sample as well.

GUSBP11 shows a reversed and less consistent pattern. The mutant samples show high PSI (0.82 - 1.00) while wildtype samples seem to be more variable (0.30 - 1.00).

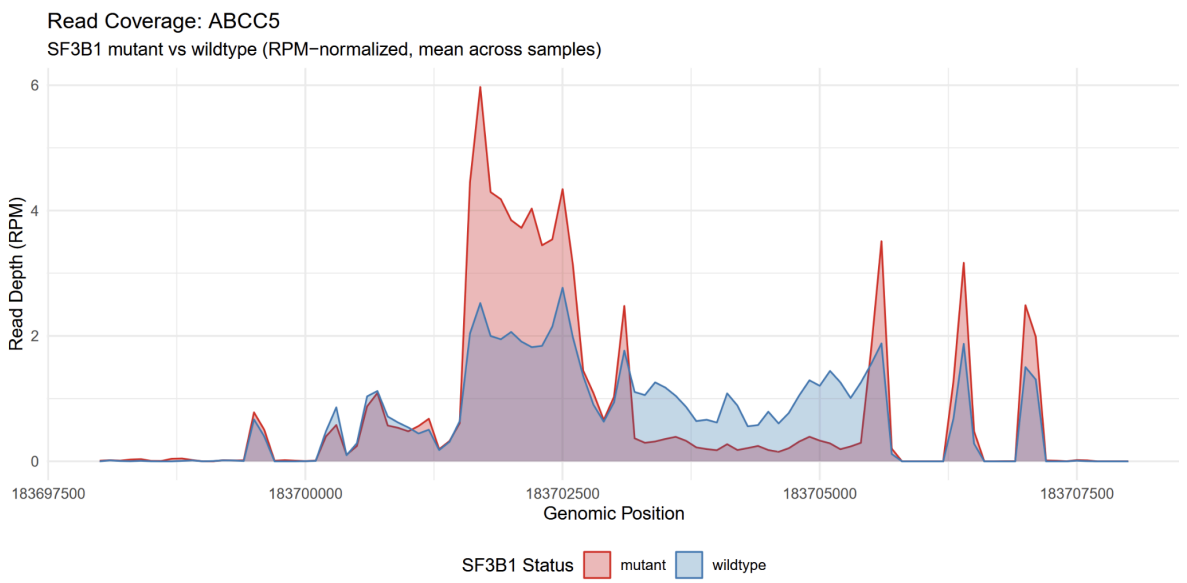
ANKHD1 shows the least consistent pattern, as mutant samples vary strongly (0.22 - 1.00) and wildtype samples also show high PSI (0.79 - 1.00). No clean separation between groups can be seen.

Conclusion:

SF3B1 mutation has a gene-specific effect on alternative splicing since it consistently suppresses inclusion in ABCC5 and CRNDE, while promoting or having variable effects on GUSBP11 and ANKHD1. This is consistent with SF3B1's known role as a splicing factor whose mutation causes widespread but non-uniform splicing dysregulation.

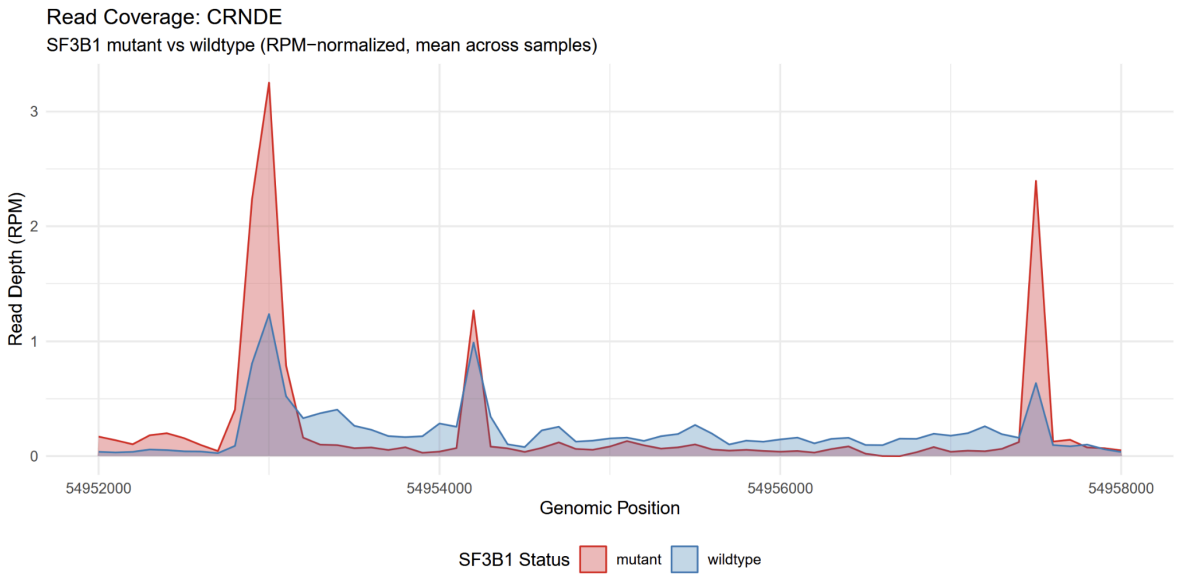
In the following section the coverage plots for the discussed genes (ABCC5, CRNDE, UQCC1) are displayed and interpreted. The x-axis in the respective plots display the genomic position during the read, and the y-axis the average read depth. The red area displays the SF3B1 mutant group, whereas the blue plot shows the wildtype.

Coverage - ABCC5



In this coverage plot the gene ABCC5 can be seen, comparing RNA-seq read depth between SF3B1 mutant and wildtype samples across a 10kb genomic region on chromosome 3 (between positions 183.697.500 - 183.708.00). The main observation is that the peaks correspond to exons (e.g. regions of reactive transcription where the reads accumulate). Further it can be noted that the red peak at around position 183.702.00 - 183.702.500 where the mutant reaches half of the average read depth of ~200 (6 RPM), while the wildtype only reaches around ~80 (2.8 RPM). This could indicate altered splicing that retains or creates a new exon in the mutant, or increased inclusion of a specific exon, or a shift in splice site usage.

Coverage - CRNDE



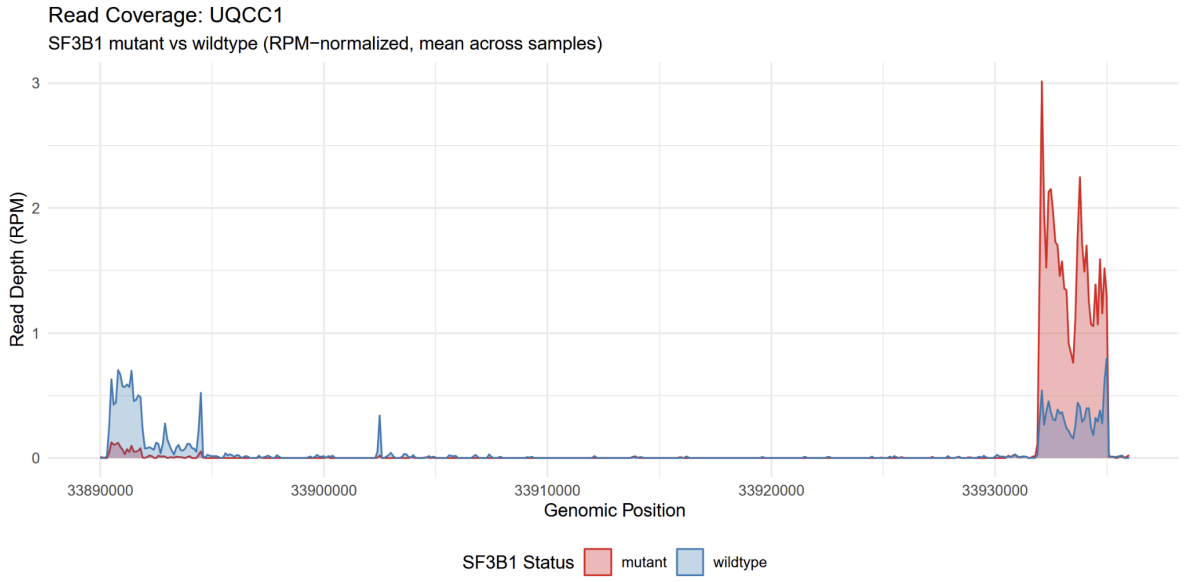
This plot shows read coverage for CRNDE,a long non-coding RNA, across a roughly 7 kb region. The overall pattern is one of consistently higher mutant coverage at specific peaks, while the baseline coverage between peaks is low in both groups.

The most prominent feature is the sharp mutant peak around position 54,952,500, where read depth reaches approximately 98 (3.4 RPM) in mutants compared to roughly 35 (1,4 RPM) in wildtype. A second peak around 54,954,000 shows a similar pattern, with mutant coverage about equal to or slightly above wildtype. The rightmost peak near 54,957,500 again shows strongly elevated mutant coverage of around 75 (2,8 RPM), while wildtype reaches only about 20 (1,3 RPM).

Notably, between the peaks the wildtype coverage is actually slightly higher than mutant in several stretches, which mirrors the pattern seen in UQCC1 and ABCC5. This reciprocal relationship between peaks and inter-peak regions is a recurring signature of SF3B1-driven splicing alterations.

The sharp, narrow shape of the mutant peaks is worth noting. Rather than broad exon-wide elevation, the coverage spikes are very focal, which could suggest the activation of very precise cryptic splice sites creating short novel exonic segments, or alternatively a shift in the exact splice site boundaries within existing exons.

Coverage - UQCC1



This graph displays the read coverage for UQCC1, again comparing SF3B1 mutant and wildtype across a genomic window of 45 kb (33.888.000 - 33.935.000).

This noticeable pattern of reduced mutant coverage upstream and strongly increased mutant coverage downstream is unusual. The most likely explanation is that SF3B1 mutation causes a shift in splicing that redirects the transcript toward a different ending. SF3B1 mutations are known to activate cryptic splice sites, and in some genes this leads to a completely different set of terminal exons being used. The mutant may produce a transcript that skips the upstream exons entirely and instead uses a downstream region, which would explain the large accumulation of reads at the right end of the plot.

Another possibility is that the right side cluster represents a cryptic exon or retained intron that is normally silenced but becomes incorporated into the transcript specifically in SF3B1 mutants.

UQCC1 plays a role in mitochondrial complex I assembly, so disrupted splicing of this gene could impair mitochondrial function. This is potentially relevant in hematologic malignancies, where SF3B1 mutations are frequently found.

5. Discussion & Pain Points

The process of reproducing a decade-old bioinformatic study highlighted several critical challenges regarding software evolution and data standards. One of the primary discussion points was the technological gap between the two original publications. Our analysis confirmed that the discrepancy in the 2013 findings was largely due to the varying sensitivity of the tools employed at the time. By implementing rMATS and a more recent genomic reference, we were able to provide a more robust statistical framework for detecting cryptic splice sites that were overlooked in the original paper.

From a technical perspective, the most significant "Pain Point" involved the configuration of the high-performance computing environment. Building the reference genome index required substantial memory resources (approximately 30GB RAM) for alignment, so instead we downloaded the pre-build GTF. Aligning large-scale RNA-seq data necessitated careful resource allocation on the server. We also had to engage the use of a temporary folder for the storage of intermediate files to stay in the allocated disk storage quota. Furthermore, ensuring full reproducibility through Nextflow and managing GitHub authentication via Personal Access Tokens (PAT) added another layer of complexity to the collaborative workflow. Finally, reconciling the data accessions (SRA/GEO) across different publications proved to be a time-consuming but essential step to ensure that our re-analysis was performed on the exact same patient populations as the original studies.

Comparison to original paper:

The key differences between the results lie in the choice of tools and general workflow approach. In this section, a breakdown of the main differences is presented.

Alignment

HISAT2 was used for read alignment in the present analysis, which is a modern splice-aware aligner. Furney et al. (2013) predated HISAT2 and most likely used TopHat2, which was the standard at the time. HISAT2 is generally considered more accurate and faster, particularly at handling novel splice junctions, and may therefore capture splicing events with higher sensitivity.

Differential splicing detection

Both DEXSeq and rMATS were employed for differential splicing detection in the present pipeline. Furney et al. used DEXSeq and MATS, an earlier version of rMATS, alongside the microarray-based tool GenoSplice EASANA. The use of the updated rMATS rather than the original MATS is methodologically relevant, as rMATS incorporates substantially improved statistical modeling and handles replicates more robustly. This difference may account for variations in PSI values or in the set of significant splicing events identified.

Annotation

The present pipeline processes the GTF annotation through DEXSeq_PREPARE_ANNOTATION and HISAT2_EXTRACT_SPLICESITES prior to analysis. Depending on the version of Ensembl or GENCODE used, the resulting exon bin definitions may differ considerably from those available to Furney et al. in 2013, as gene models have been substantially updated in the intervening years. This could affect which exon bins are counted and how PSI values are calculated, particularly for genes such as UQCC1 whose annotation has since been revised.

Sample size and composition

Furney et al. performed RNA-seq analysis on only 3 SF3B1-mutant and 9 wildtype uveal melanoma tumors. The present dataset comprises 8 samples across mutant and wildtype groups and appears to originate from a different cancer type. Differences in statistical power and tumor biology between cancer types will naturally produce variation in effect sizes and PSI values, even when the same target genes are examined.

Coverage plot normalization

Normalization strategy represents another potential source of quantitative differences. Furney et al. normalized coverage to the total number of mapped bases per gene, whereas the present pipeline may apply a different strategy such as reads per million or per-sample mean depth. Such differences would not alter the direction of the observed splicing changes, but would affect absolute read depths and the visual contrast between groups.

Conclusion

The core finding that ABCC5, UQCC1, and CRNDE are differentially spliced in SF3B1-mutant samples is consistently replicated across both analytical approaches, which strengthens the biological validity of the result. Quantitative differences in PSI values and read depths are most likely attributable to the use of updated tools, a more recent genome annotation, and a distinct cancer type in the present dataset, rather than to any methodological error in either analysis.

6. References

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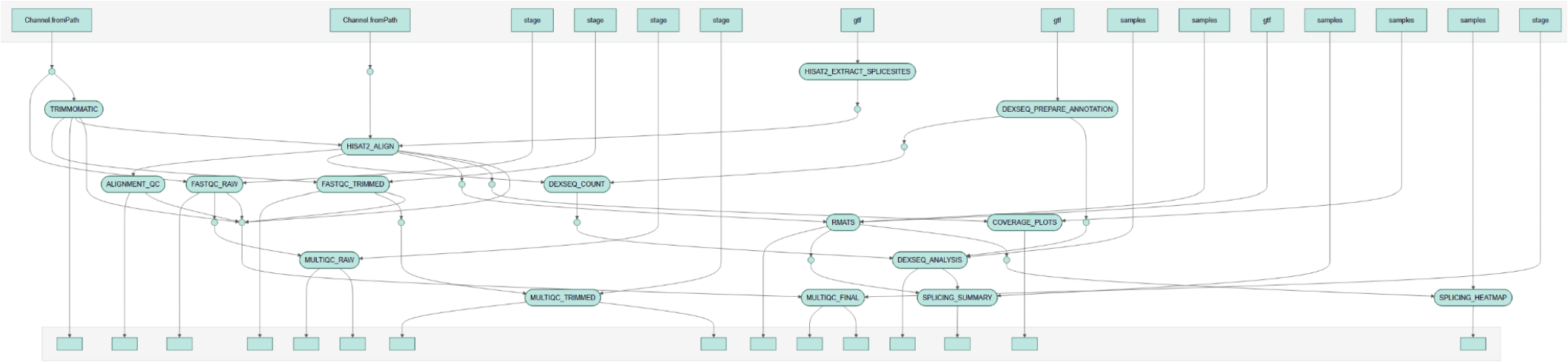
APPENDIX



Graph - Nextflow Workflow

Explanation of graph

Inputs (top row)

The rectangular boxes at the top are input channels, including raw FASTQ files via **Channel.fromPath**, a GTF annotation file, and sample metadata files (**samples**), as well as **stage** inputs feeding into the next steps.

Quality control and trimming (left side)

The pipeline starts with **TRIMMOMATIC**, which trims raw reads for adapter sequences and quality. This feeds into **FASTQC_RAW** and **FASTQC_TRIMMED** for quality assessment before and after trimming, and then into **MULTIQC_RAW** and **MULTIQC_TRIMMED** to aggregate those QC reports into summary HTML files.

Alignment (center-left)

HISAT2_ALIGN takes the trimmed reads and aligns them to a reference genome. The GTF is first processed by **HISAT2_EXTRACT_SPLICESITES** to provide splice-aware alignment information. **ALIGNMENT_QC** is run on the resulting BAM files.

Differential exon usage (center)

DEXSEQ_PREPARE_ANNOTATION processes the GTF file to prepare it for DEXSeq, and **DEXSEQ_COUNT** quantifies reads per exon bin. **DEXSEQ_ANALYSIS** performs the statistical testing for differential exon usage between SF3B1 mutant and wildtype groups.

Alternative splicing (center-right)

RMATS runs in parallel to perform splicing event analysis, detecting alternative splicing events such as skipped exons, retained introns, and alternative splice site usage.

Visualization and summary outputs (right side)

COVERAGE_PLOTS generates the read coverage plots that have been analyzed for ABCC5, UQCC1, and CRNDE. **SPLICING_SUMMARY** collates the results,

and **SPLICING_HEATMAP** produces the PSI heatmap. **MULTIQC_FINAL** aggregates the final alignment and processing QC metrics.

Outputs (bottom row)

The boxes at the bottom represent the final output files saved to their respective directories from each step.